

Date	16 July 2026
Project Name	Smart Lender AI: An Intelligent Machine Learning System for Loan Approval Prediction
Student Name	Jinkala Sunitha
Maximum Marks	3 Marks

REQUIREMENT ANALYSIS: CUSTOMER JOURNEY MAP

This Customer Journey Map traces the end-to-end lifecycle of our user persona, **The Loan Applicant**, as they navigate the credit evaluation process. By mapping out sequential touchpoints, system requirements are translated directly from emotional bottlenecks into targeted interface features and algorithmic processes.

STAGE	1. DISCOVERY & ENTRY	2. APPLICATION SUBMISSION	3. SYSTEM PROCESSING	4. EVALUATION & RESULTS
User Actions	- Visits the Smart Lender AI platform landing page. - Reviews the required input parameters.	- Enters demographics, assets, and co-applicant details. - Submits the structured form securely.	- Waits for backend operations to finish. - Checks status on screen.	- Views instant decision output (Approved/Rejected). - Examines feature feedback dashboards.
Touchpoints	Web Portal Landing Page, Form Description Box	Dynamic Input Form, Input Verification Fields	Model Processing Pipeline, API Endpoint	Output Interface, Analytics Charts
User Thoughts	"I hope this system is straightforward and doesn't ask for tedious paper uploads."	"Is my financial data safe? Are these fields clearly explaining what they mean?"	"How long will this processing spin? Will it throw a generic system timeout?"	"The verdict is immediate! Why did it decide this way, and what factors were key?"
Customer Emotions	Curious / Apprehensive	Anxious / Attentive	Impatient / Suspicious	Relieved / Informed
Pain Points	Complicated layouts, confusing acronyms, lack of guidance.	Session timeouts, form validation errors, too many inputs.	Lack of system feedback, long loading delays, "black-box" feel.	Ambiguous "Denied" labels without clear explanation or reasoning.
System Requirements	REQ-01: Intuitive UI Layout Clean dynamic landing page	REQ-02: Structured Validation Dynamic type checking to prevent incorrect range values or letters.	REQ-03: Real-Time API Under-1-second response times	REQ-04: Analytical Transparency Provide automated explanations

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	requesting only crucial variables[cite: 10, 11].		using optimized Flask routing[cite: 10].	mapping results directly to the prediction[cite: 10, 11].